

EXPERIMENT 5

DATA VISUALIZATION USING MATPLOTLIB

Aim : To create and analyze various data visualizations such as Line Charts, Bar Charts, Pie Charts, Histograms, Scatter Plots, and Box Plots using Matplotlib for effective data interpretation.

Algorithm :

1. Import the required libraries and load the Instagram Analytics dataset.
2. Select the required columns for visualization.
3. Prepare or group the data according to the visualization.
4. Choose the appropriate chart type for the selected data.
5. Create the chart using the required Matplotlib function.
6. Add a suitable title, axis labels, and legend if required.
7. Display the visualization and interpret the results.

Program :

```
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Load the Instagram Analytics dataset
df = pd.read_csv("../Instagram_Analytics.csv")

# Display the first five records
print("===== FIRST FIVE RECORDS =====")
display(df.head())
```

===== FIRST FIVE RECORDS =====

	post_id	account_id	account_type	follower_count	media_type	content_category	traffic_source	has_call_to_action	post_datetime
0	IG0000001	7	brand	3551	reel	Technology	Home Feed	1	2024-11-30 06:00:00
1	IG0000002	20	creator	31095	image	Fitness	Hashtags	1	2025-08-15 15:00:00
2	IG0000003	15	brand	8167	reel	Beauty	Reels Feed	0	2025-09-1 16:00:00
3	IG0000004	11	creator	9044	carousel	Music	External	0	2025-09-15 03:00:00
4	IG0000005	8	creator	15986	reel	Technology	Profile	0	2025-03-2 09:00:00

===== FIRST FIVE RECORDS =====

```

# Check the shape of the dataset
print("==== SHAPE OF DATASET =====")
print(df.shape)
# Check the data types of all columns
print("==== DATA TYPES =====")
print(df.dtypes)
# Display descriptive statistics for numerical columns
print("==== DESCRIPTIVE STATISTICS =====")
display(df.describe())

```

```

==== SHAPE OF DATASET =====
(29999, 23)
==== DATA TYPES =====
post_id                object
account_id             int64
account_type           object
follower_count         int64
media_type             object
content_category       object
traffic_source         object
has_call_to_action     int64
post_datetime          object
post_date              object
post_hour              int64
day_of_week            object
likes                 int64
comments              int64
shares                int64
saves                 int64
reach                 int64
impressions            int64
engagement_rate        float64
followers_gained       int64
caption_length         int64
hashtags_count         int64
performance_bucket_label object
dtype: object

```

```

daily_engagement = df.groupby(
    "post_date"
)["engagement_rate"].mean()

plt.figure(figsize=(10,5))

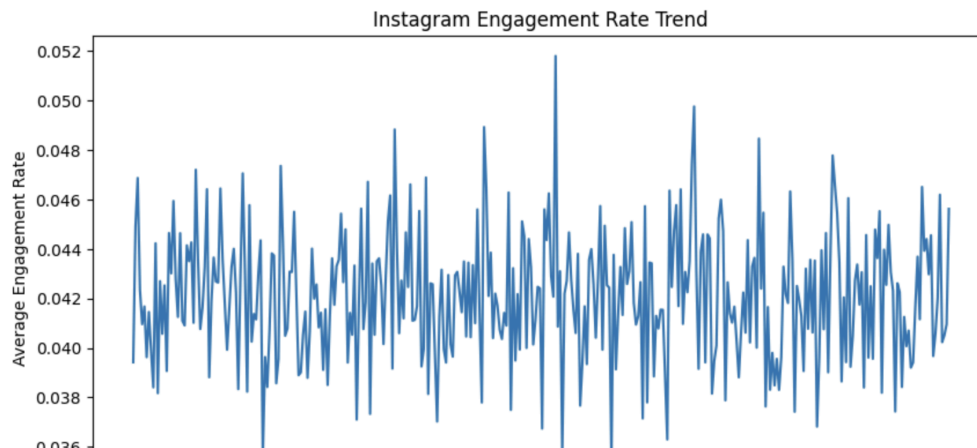
plt.plot(
    daily_engagement.index,
    daily_engagement.values
)

plt.title("Instagram Engagement Rate Trend")
plt.xlabel("Post Date")
plt.ylabel("Average Engagement Rate")

plt.xticks(rotation=45)

plt.show()

```



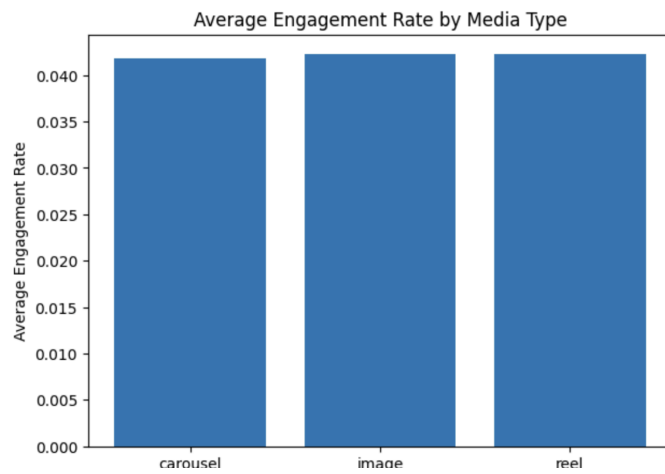
```
media_engagement = df.groupby(
    "media_type"
)["engagement_rate"].mean()

plt.figure(figsize=(7,5))

plt.bar(
    media_engagement.index,
    media_engagement.values
)

plt.title("Average Engagement Rate by Media Type")
plt.xlabel("Media Type")
plt.ylabel("Average Engagement Rate")

plt.show()
```

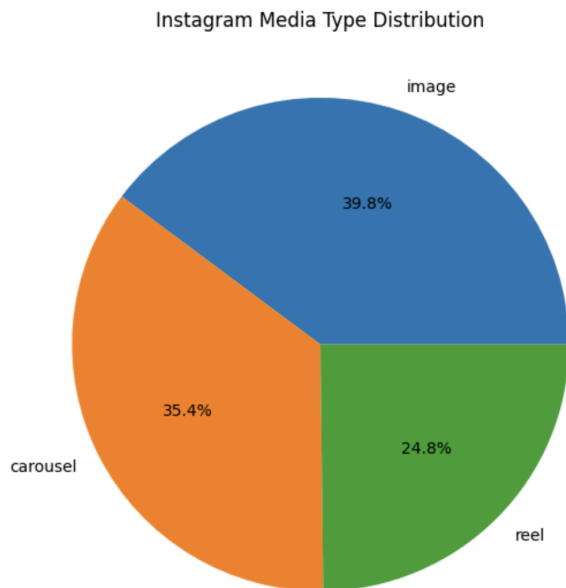


```
plt.figure(figsize=(7,7))

plt.pie(
    media_counts,
    labels=media_counts.index,
    autopct="%1.1f%%"
)

plt.title("Instagram Media Type Distribution")

plt.show()
```

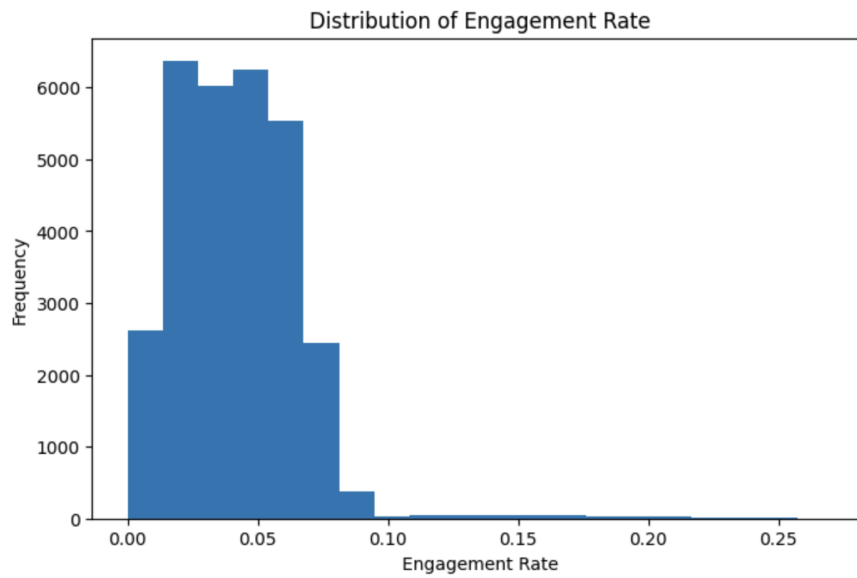


```
plt.figure(figsize=(8,5))

plt.hist(
    df["engagement_rate"],
    bins=20
)

plt.title("Distribution of Engagement Rate")
plt.xlabel("Engagement Rate")
plt.ylabel("Frequency")

plt.show()
```

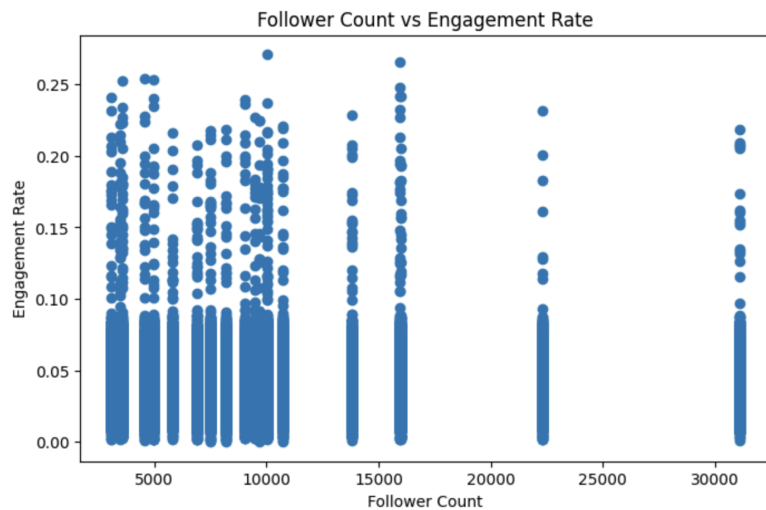


```
plt.figure(figsize=(8,5))

plt.scatter(
    df["follower_count"],
    df["engagement_rate"]
)

plt.title("Follower Count vs Engagement Rate")
plt.xlabel("Follower Count")
plt.ylabel("Engagement Rate")

plt.show()
```

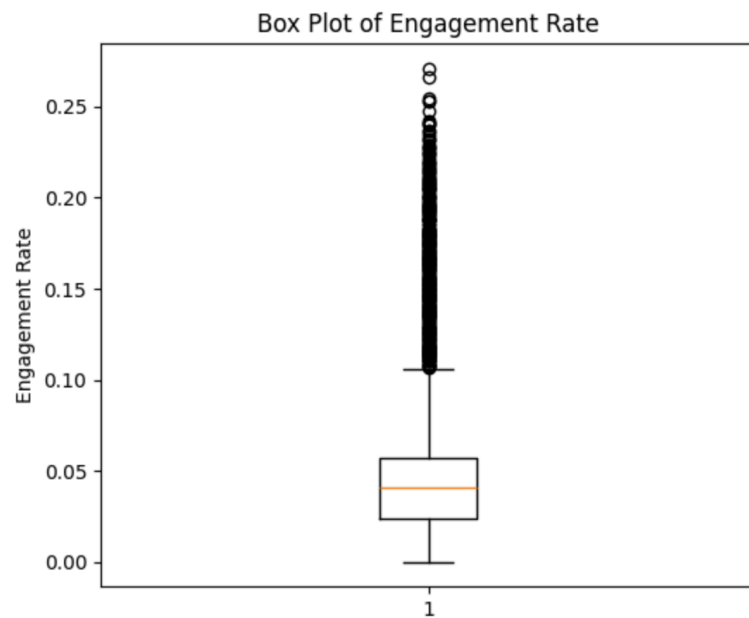


```
plt.figure(figsize=(6,5))

plt.boxplot(
    df["engagement_rate"]
)

plt.title("Box Plot of Engagement Rate")
plt.ylabel("Engagement Rate")

plt.show()
```



```
plt.title("Reach vs Impressions Over Time")
plt.xlabel("Post Date")
plt.ylabel("Average Count")

plt.legend()

plt.xticks(rotation=45)

plt.show()
```



